JMM Special Session Proposal

Clare Kennedy, Darcy, Miller

1 Title and Organizers

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2 Public Description

The broad idea of finding "shape" in data has led to insights in various fields, including neuroscience and biomedical research. A recent example is researchers were able to identify novel subtypes of diabetes (citation) and breast cancer using the Mapper algorithm, which transforms data into a simplicial complex. In this session, speakers will discuss their real-world use of topological ideas and frameworks, as well as dicuss the mathematical tools and theory used in topological data analysis.

3 Sample Speaker List

- Brad Theilman (Sandia National Laboratories)
- Enrique Alvarado (Iowa State University)
- Aurora Clark (Utah Chemistry)
- James Melville
- Elizabeth Munch
- Brittany Story
- Mathieu Carriere

4 Long-Form Description

This session will bring together speakers from many disciplines, under the umbrella of topological data analysis (TDA) and visualization. This will include mathematicians who study algorithms such as Mapper and concepts like persistent homology and dimension reduction, as well as researchers who are not mathematicians who have used these ideas successfully in their work. There will be ample discussion of software and opportunities to collaborate on larger TDA and visualization libraries, an area currently lacking in availability, user friendliness, and standardization.

Mapper in particular has seen a variety of applications in various domains, such as neuroscience, clinical medicine, genomics, and sociology, each with their own approach to parameter selection and analysis. Despite this growing body of work, the field lacks a standardized methodology for parameter selection. Some work has been done on optimizing filter functions (citation) and covers (citation), but their use for researchers in practice is necessarily limited, as the software landscape for Mapper is fragmented; implementations differ in their supported data formats, scalability, library APIs, and visualization capabilities. This leaves researchers unaware of best practices or critical technical details from other fields, which can lead to suboptimal, misleading, or even invalid analyses. The problem is not unique to Mapper; software for TDA and visualization is commonly out-of-date, unorganized, or closed-source.

By bringing together mathematicians and researchers at the session, we aim to bridge some of the gaps that exist between theory and practice, and foster new and exciting opportunities for further collaboration. The interdisciplinary nature of the session will bolster this, ideally attracting JMM attendees who might not otherwise have had the opportunity to see or work on such projects. Especially as the world becomes more and more saturated with data in the age of information and AI, there is more need than ever for mathematicians to make sense of complex, often high-dimensional data. TDA and visualization research enables both these mathematicians and those from other fields to make new connections and discoveries, and sessions such as these help keep those discoveries informed, accurate, and forward-focusing.

5 MSC

One of:

- 101: Teaching and Learning
- 102: Recreational Mathematics
- 103: Professional Development and Professional Concerns
- 104: Wider Issues